Week 9, Lecture 1: From Optimization to MCMC

Instructor notes. Part 1: prep reading. Part 2: podium cheat sheet

How to prepare (for students)

- **Big picture.** Last Wednesday we opened up `optim()` and saw how Newton's method uses the slope and curvature of the NLL to step toward the MLE. Today: two search methods that need no derivatives (Nelder-Mead and simulated annealing), then a small change to simulated annealing that turns it from a way to *find the best* parameter values into a way to *sample the whole posterior distribution*. That method, MCMC, is how most Bayesian models are fit.
- **Read:** Bolker 7.2.3.2 and 7.2.4 (Nelder-Mead and simulated annealing, printed p. 302-310) and 7.3 through 7.3.2 (MCMC, burn-in, convergence, p. 310-315).
- **Review:** Bayes' rule for parameters (posterior $\propto$ likelihood $\times$ prior) and the Beta-binomial germination example from Week 8; the log rule $\log(a/b) = \log a - \log b$.
- **Tuesday's lab** has you build and test your own MCMC sampler, so bring questions from today.

Overview (instructor only)

Beat	Mode	Time
1. Big picture	BOARD	~4 min
2. Nelder-Mead	BOARD $\rightarrow$ R	~7 min
3. Simulated annealing	BOARD $\rightarrow$ R	~8 min
4. Metropolis	BOARD $\rightarrow$ R	~15 min
5. Wrap	BOARD	~2 min
Discussion	R (reveal)	~6 min

No WOD today, to leave room so Tuesday's lab can start without a long lecture. **BOARD** = whiteboard; **R** = switch to `CODE_Week9_MCMCAndStan.R` (Monday section; numbers match). The script holds only the R demos; figures in these notes are board references only.

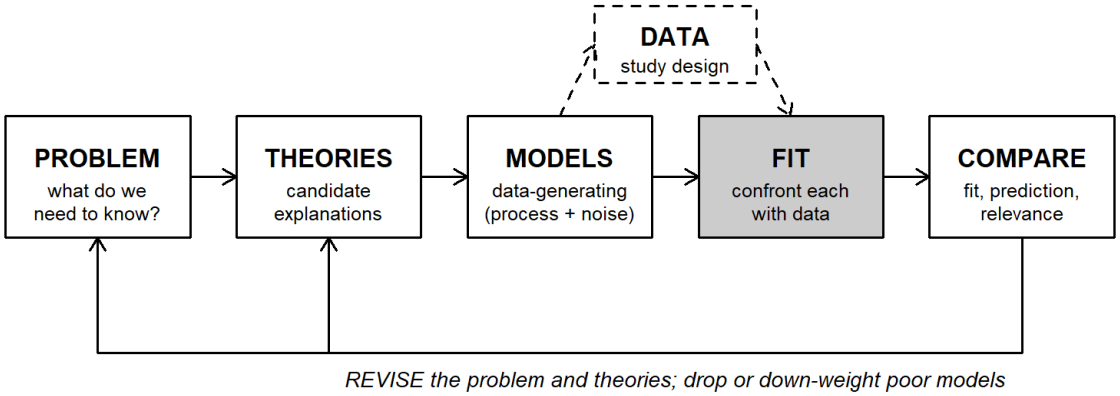
Where this came from. Sections 2-3 are the Nelder-Mead and simulated annealing material planned for Week 8 Lecture 2 (Oct. 14), which only got through Newton's method. Both are trimmed to their core ideas. The shape-scale correlation pause and the reparameterization "if time" material from Week 8 become Lab 7 Act 2's centering step, where removing a strong

parameter correlation fixes a badly mixing chain (the same reason the myxo banana mixes slowly).

If running behind, skip the SANN pause in section 3 and the two-parameter myxo chain in section 4. Don't cut the discussion: it sets up Lab 7.

Part 1: Lecture content

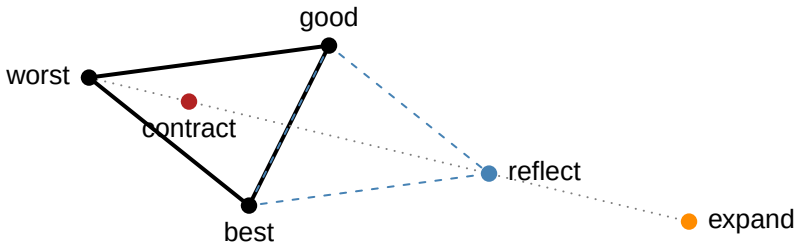
1. Big picture (~4 min) [BOARD]



- Workflow on the board (a quick version of the above); point at **FIT**.
- Recap: fitting = finding the bottom of the NLL surface. Newton and BFGS use slopes; fast near the answer, can fail far from it.
- Today, two questions: (1) how do we search when slopes are unavailable or misleading? (2) For Bayesian fits, how do we get the posterior when there are too many parameters for a grid and no conjugate shortcut?
- The answer to (2) grows out of the answer to (1).

2. Nelder-Mead (~7 min) [BOARD → R]

Board. Sketch one triangle (three points for two parameters) and its moves:



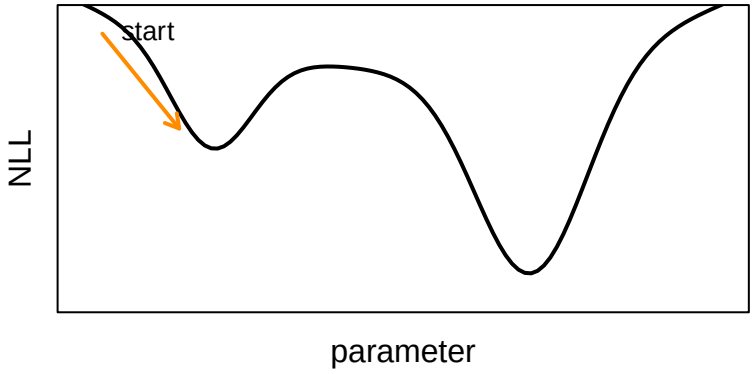
- Reflect the worst point through the midpoint of the other two. Best yet? Expand. Still bad? Contract. Nothing works? Shrink toward the best point.
- No derivatives: only compares NLL values.

R demo (myxo Gamma surface from Week 8). The triangle expands across the contours, contracts as it turns into the valley, then crawls along the valley floor. `optim()`'s default is Nelder-Mead: 147 NLL evaluations, NLL 37.667.

Point to land. Slower than BFGS, but bumps, kinks, and thresholds in the NLL don't fool it.

3. Simulated annealing (~8 min) [BOARD → R]

Board. A curve with two valleys (Bolker's Figure 7.1). Any downhill method finds whichever valley it starts in.



The rule (Bolker p. 306):

1. Propose a random nearby point.
2. Lower NLL: move there.
3. Higher by Δ: move there anyway with probability $e^{-\Delta/T}$.
4. Repeat; lower the "temperature" T now and then.

Hot: accepts most uphill steps and explores. Cold: almost pure downhill. Start hot, cool slowly.

R demo. Acceptance table, then the annealer on the myxo surface: it finds the valley, wanders along it, settles as T falls. About 3,000 evaluations for what Nelder-Mead did in 147.

Pause (~1 min, skippable). `optim(..., method = "SANN")` with defaults gives NLL 40.77 after 10,000 evaluations and `convergence = 0` (Nelder-Mead: 37.67). Broken? *Land:* untuned, not broken. SANN always reports 0. Use a global search to find the right valley, then polish with Nelder-Mead or BFGS.

4. Metropolis (~15 min) [BOARD → R]

Key idea. Keep the annealing rule but change three things, and the chain of visited points becomes a sample from the posterior.

Board, in three steps.

1. Fix $T = 1$ and never cool. The acceptance probability for an uphill step becomes $e^{-(\text{NLL}_{new} - \text{NLL}_{old})} = \mathcal{L}(\theta_{new})/\mathcal{L}(\theta_{old})$.
2. Swap the likelihood for the posterior:

$$P(\text{accept}) = \min\left(1, \frac{P(\theta_{new} \mid y)}{P(\theta_{old} \mid y)}\right)$$

3. Keep **every** step, not only the best one.

Then write Bayes' rule into the ratio:

$$\frac{P(\theta_{new} \mid y)}{P(\theta_{old} \mid y)} = \frac{\mathcal{L}(\theta_{new})\, P(\theta_{new})/P(y)}{\mathcal{L}(\theta_{old})\, P(\theta_{old})/P(y)}$$

$P(y)$ cancels. Computing it is why Week 7 used a grid and Week 8 relied on conjugacy; neither works with many parameters or arbitrary priors. Metropolis never needs it.

Why it lands on the posterior (board, 1 min). Two values, A and B, with $P(A \mid y) = 2\,P(B \mid y)$. From B, a proposal to A is always accepted; from A, a proposal to B is accepted half the time. In the long run the chain sits at A twice as often as at B. Same logic everywhere: time spent near $\theta \propto$ posterior.

Sticking point. A rejected proposal means *record the current value again.* Students often skip rejected steps; that throws away the extra weight the chain puts on high-posterior regions. Related: the chain is not supposed to converge to a point. Wandering is the goal.

R demo.

- Germination (30 of 40, flat prior; exact posterior Beta(31, 11)). Start at 0.5, proposal sd 0.1, 5,000 steps. Acceptance 59%. Chain mean 0.734 vs. exact 0.738; 95% interval 0.59-0.85 vs. exact 0.60-0.86. The histogram matches the curve we already know.
- Code note: compute log posteriors as sums of logs, never a raw product (27 densities multiplied together can underflow).
- **Pause (~1 min).** `anneal()` accepted with `runif(1) < exp(-delta / temp)`; `metropolis()` uses `log(runif(1)) < lp_prop - lp_current`. Different rules? *Land:* no. $u < e^x \iff \log u < x$, and with $T = 1$ and the NLL difference replaced by the log posterior difference, the two lines make the same decision. Only a *difference* is exponentiated, so neither can underflow.
- Myxo Gamma, two parameters, flat priors, 20,000 steps: the cloud of points fills the banana. 95% credible interval for shape: 29 to 81 (first 2,000 steps dropped; say why in one line, and leave burn-in for the discussion). This chain mixes slowly along the banana (other step sizes shift the lower bound by several units); Lab 7 explains why.

5. Wrap (~2 min) [BOARD]

- Point at **FIT**. Grid, Newton/BFGS, Nelder-Mead, annealing: ways to find the minimum. Metropolis: a way to sample the posterior.
- Same engine (random proposal + acceptance rule); different goal.
- Tomorrow: you build one and learn how to tell whether to trust it.

Discussion (~6 min) [R reveal]

> Three labmates run Metropolis on the germination data (30 of 40, flat prior; exact posterior mean 0.738). Same start ($p = 0.05$), 1,000 steps each, different proposal sd:
>
> | proposal sd | accepted | chain mean |
> |---|---|---|
> | 0.005 | 87% | 0.38 |
> | 0.1 | 59% | 0.73 |
> | 3 | 4% | 0.70 |
>
> The first labmate says theirs is best, since nearly every proposal was accepted. Which chain would you trust? What would you check or change?

After a few minutes, show the three trace plots (script, Discussion section).

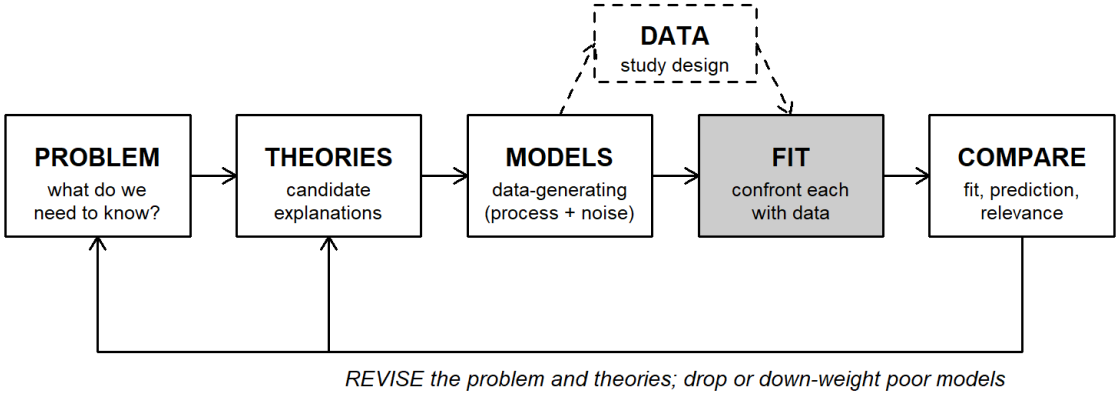
Where to land. Tiny steps are almost always accepted but barely move: after 1,000 steps that chain is still climbing from 0.05 (it never finished burn-in). Huge steps are almost always rejected, so the chain sits still for long stretches; its mean is close by luck, from a few dozen distinct values. The middle chain mixes well. A high acceptance rate is not a quality score. Look at the trace plot, drop the burn-in, and tune the proposal width. Lab 7 picks up from here.

PART 2: PODIUM CHEAT SHEET

Terse on purpose. Detail: Part 1. R = Monday section of CODE_Week9_MCMCAndStan.R.

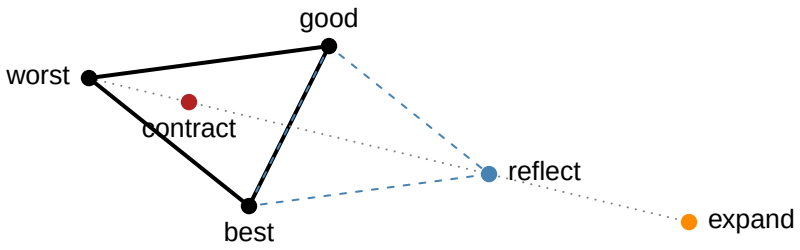
No WOD. Lecture ~36 min, discussion ~6 min. Behind? Skip SANN pause and myxo chain.

1. Big picture (~4 min) [BOARD]



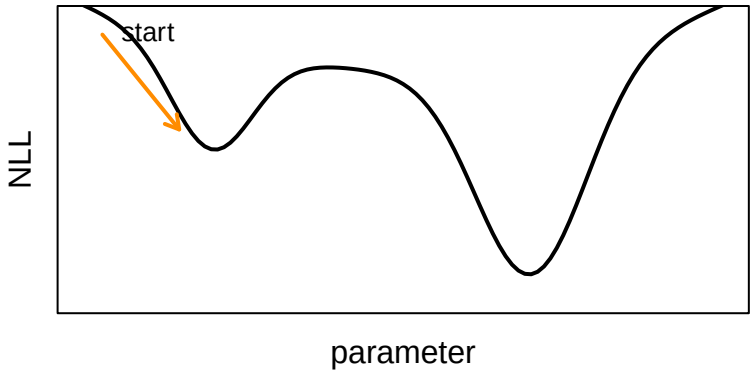
- Workflow on board; point at **FIT**
- Newton/BFGS: slopes; fast near answer, fail far away
- Q1: search without slopes?
- Q2: posterior with too many params for a grid, no conjugacy?
- Q2's answer grows out of Q1's

2. Nelder-Mead (~7 min) [BOARD → R]



- Reflect worst; best yet → expand; still bad → contract; else shrink
- Compares NLL values only, no derivatives
- **R:** triangles on myxo surface; `optim()` default, 147 evals
- Slower than BFGS; not fooled by bumps or kinks

3. Simulated annealing (~8 min) [BOARD → R]



- Downhill methods find the nearest valley
- Random step; worse by Δ: accept w.p. $e^{-\Delta/T}$; cool T
- Hot explores, cold settles
- **R:** acceptance table; annealer path; ~3,000 evals vs. 147

Pause (skippable): SANN: NLL 40.77, conv 0. Broken?

- Untuned; always 0. Find valley, then polish with NM/BFGS

4. Metropolis (~15 min) [BOARD → R]

1. $T = 1$: $e^{-\Delta \text{NLL}} = \mathcal{L}_{new}/\mathcal{L}_{old}$
2. Posterior instead: accept w.p. $\min\left(1, \frac{P(\theta_{new}|y)}{P(\theta_{old}|y)}\right)$
3. Keep **every** step

- Write Bayes in the ratio: $P(y)$ **cancels**
- A vs. B, A twice as probable: B→A always, A→B half ⇒ 2:1 time
- **Rejected = record current value again**
- Not converging to a point; wandering is the goal

R:

- Germination: mean 0.734 vs. 0.738; accept 59%
- Log posterior = sum of logs (a product underflows)
- **Pause:** `runif(1) < exp(-delta / temp)` vs. `log(runif(1)) < lp_prop - lp_current`? Same rule ($u < e^x \iff \log u < x$)
- Myxo: cloud fills banana; shape 29-81

5. Wrap (~2 min) [BOARD]

- **FIT**: optimizers find the minimum; Metropolis samples the posterior
- Same engine, different goal
- Tomorrow: build one, learn when to trust it

Discussion (~6 min) [R reveal]

Start 0.05, 1,000 steps; exact mean 0.738:

- sd 0.005: 87% accepted, mean 0.38
- sd 0.1: 59%, mean 0.73
- sd 3: 4%, mean 0.70
- "First is best: nearly all accepted." Trust which?

Then show trace plots.

- Tiny steps: never finished burn-in
- Huge steps: stuck; close by luck
- Acceptance $\neq$ quality; trace plot, burn-in, tune width $\rightarrow$ Lab 7